MANASWI RAGHURAMA (PhD)

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WORK EXPERIENCE

Jul, 2026 – now: Postdoctoral Researcher (remote consultant), Missouri Botanical Garden, St. Louis, USA:

- Working with the SNAPP (Science for Nature and People Partnership)-MoRe (Monitoring Restoration) working group (https://snapp-more.github.io/) on simplifying the monitoring of restoration outcomes in tropical forests, savannas, and mangroves.
- Working on three inter-related projects:
    - Synthesizing long-term monitoring datasets to evaluate whether short-term indicators can reliably predict long-term ecological restoration outcomes.
    - Review of monitoring standards, protocols, frameworks, guidelines, etc., to build an online tool where practitioners can find potential monitoring resources of interest that they can use.
    - Conducting an online questionnaire to understand how restoration practitioners are currently monitoring the outcomes of their projects in the tropics.

May, 2025 – Jun, 2026: Postdoctoral Associate, University of Minnesota, Twin Cities, USA:

- Started working with the SNAPP-MoRe working group (see above for project details).

Jul, 2024 – Jan, 2025: Consultant, Centre for Pastoralism:

- Synthesizing a policy report using survey data from pastoral communities regarding dung economy, sheep penning, and pastoral behavior from two states of central India.

Nov, 2015 – Jan, 2016: Research Fellow, Panchabhuta Conservation Foundation, Karnataka, India:

- Conducted social science research in the Aghanashini river estuarine area, exploring how different communities use natural resources in the area. Particularly focused on water, agroecological products, and medicinal plants.

PhD RESEARCH EXPERIENCE

2016-2024: National Centre for Biological Sciences, Bengaluru, India: PhD - completed

- Thesis: Community and Ecosystem impacts of woody plant invasion in the Shola-Grasslands of the Nilgiris
- Studied impacts of Nitrogen-fixing woody invasive plants on the vegetation community, soil nitrogen cycling, and the export of nutrients and sediments, in the montane grasslands of southern India.
- Used insights from this work to build a conceptual framework to guide the restoration of forest-grassland mosaics invaded by woody exotics.
- Used variety of statistical tools and techniques: regression analysis, generalized linear models, mixed models, zero-inflated models, hurdle models, additive mixed models, multivariate analysis.
- Two peer-reviewed publications; two in preparation; one policy report on the restoration of terrestrial ecosystems in India (details below).
- Grant writing – awarded National Geographic Society's Early Career Grant for fieldwork.

PUBLICATIONS

Raghurama M, Sankaran M (2021) Restoring tropical forest–grassland mosaics invaded by woody exotics. *Restoration Ecology* 29(8):e13491. https://doi.org/10.1111/rec.13491

Raghurama M, Sankaran M (2022) Invasive nitrogen-fixing plants increase nitrogen availability and cycling rates in a montane tropical grassland. *Plant Ecology* 223(1):13-26. https://doi.org/10.1007/s11258-021-01188-4

Raghurama M, Dutta S, Nair T, Sankaran M, *et al.* (2023) Restoring India's Terrestrial Ecosystems - Needs, Challenges, and Policy Recommendations. *Zenodo*. https://doi.org/10.5281/zenodo.10040431

Das A, Niphadkar M, Nair T, Gavai P, Kelkar N, Shah MA, Hiremath A, Ramaswami G, Sundaram B, Bunyan M, Krishnaswamy J, Ramakrishnan U, Sankaran M, Raghurama M *et al.* (2026). Understanding and Managing Invasive Alien Species in India. *Zenodo*. https://doi.org/10.5281/zenodo.18911827

Raghurama M, Krishnaswamy J, Bhalla RS, Vaidyanathan S, Sankaran M (2026) Data and code: Invasive N-fixing trees transform nitrate export from being supply- to transport-limited in a forest-grassland mosaic. *Zenodo*. https://doi.org/10.5281/zenodo.19862172

Raghurama M, Krishnaswamy J, Bhalla RS, Vaidyanathan S, Sankaran M. Invasive N-fixing trees transform nitrate export from being supply- to transport-limited in a forest-grassland mosaic. (*Under revision)*

Raghurama M, Sankaran M. Invasive and exotic woody plants alter vegetation functional composition and increase vegetation homogeneity in a montane tropical grassland. *(in preparation)*

EDUCATIONAL QUALIFICATIONS

2014-2015: Trinity College, Dublin, Ireland: M.Sc. Environmental Science

- Pursued courses on environmental management, statistics, and GIS for mapping.
- Thesis: Changes in Macroalgal C:N:P ratios and Nutrient cycling by intertidal sediments with increasing Nitrogen pollution – A study in Kinvarra Bay, Ireland
    - Measured nutrient stoichiometry of macroalgae in a Nitrogen-enriched bay, and predicted how sediment-water nutrient fluxes might change due to changes in nutrient stoichiometry of organic matter.

2010-2014: Birla Institute of Technology & Science (BITS), Pilani, India: Integrated M.Sc. (Hons.) Biological Sciences

- Pursued several courses in Mathematics and Statistics.
- Thesis: Variation in the sequestration of Carbon and Nitrogen in soils with change in Vegetation and Depth: A study in the Nilgiri Mountains, India
    - Measured Carbon:Nitrogen ratio of soils of four different land cover types (grasslands, forests, plantations) in a protected area in the Nilgiris.

AWARDS AND HONORS

- **NCBS Intramural travel grant** (62,000 INR) to attend the "18th Savanna Science Network

Meeting" in Kruger National Park, South Africa in March 2020.

- **National Geographic Society's Early Career Grant** ($6,300) to fund PhD field work.

- **Government of Ireland Scholar** (Full tuition fees + €10,000 Stipend) for the MSc Environmental Science program at Trinity College, Dublin.

- **State Topper in Mathematics, International Assessments for Indian Schools** (IAIS) conducted by Educational Assessment Australia (EAA) and Macmillan Publishers India Ltd.

TEACHING AND EDITORIAL RESPONSIBILITIES

- Co-Reviewer, New Phytologist (2022)

- Teaching Assistant, "Biostatistics" Course, National Centre for Biological Sciences (2022)

CONFERENCES

Aug, 2026: 15th European Conference on Ecological Restoration (SERE (Society for Ecological Restoration - Europe)- 2026), Brest, France:

- Presented talk on "Urgent need to simplify restoration monitoring protocols in the tropics to track project goals better" in the symposium on "Integrating best practice guidelines, community engagement and digital tools for measuring restoration outcomes in mangrove restoration"

Sep-Oct, 2025: 11th World Conference on Ecological Restoration (SER2025), Denver, USA:

- Presented talk on "SNAPP-MORE: Using longitudinal data to predict how long should restoration outcomes be monitored for in the tropics" in the symposium on "Monitoring Tropical Forest Restoration from on the Ground Measurements to New Technologies".

Jul, 2024: 4th International Young Researchers Conference on Invasive Species (Online Conference):

- Presented talk on "Community and ecosystem impacts of woody plant invasion in the Shola-Grasslands of the Nilgiris".

Jul, 2023: 59th Annual Meeting of the ATBC (Association for Tropical Biology and Conservation).

- Organized symposium on "Restoring Terrestrial Ecosystems in India: Frameworks, Synthesis and the Way Forward".
- Presented talk on "Restoring India's terrestrial ecosystems: Overview for practitioners and inputs for policy makers".

Jun, 2021: 8th Global Nitrogen Conference (Virtual Meeting) held from Berlin, Germany

- Presented talks on (1) "Impacts of invasive plants on Nitrogen cycling in a montane tropical grassland" and (2) "Effects of vegetation structure on nutrient outflows from a montane tropical Forest-grassland mosaic".

Mar, 2020: 18th International Savanna Science Network Meeting, Kruger National Park, South Africa

- Presented talk on "Impacts of Nitrogen-fixing invasive plants on nutrient cycling and outflow in a tropical montane forest-grassland mosaic in the Nilgiris Biosphere Reserve, India".

Sep, 2016: Student Conference on Conservation Science, Bengaluru, India

- Presented talk on "Impact of Nitrogen pollution on macroalgal C:N:P and nutrient cycling by intertidal sediments: An experimental study in Kinvarra Bay, Ireland"

VOLUNTEERING EXPERIENCES

Apr, 2020 – Jun, 2020: Covid-19 Testing Center, National Centre for Biological Sciences, Karnataka, India:

- Administrative Manager in testing center – one of only two such centers in the state of Karnataka in the initial months of Covid-19 outbreak in India.
- Managing database of incoming samples and test results; Reporting to hospitals, block and district health officers; Managing volunteer teams.

Jul, 2014 – Apr, 2017: Co-Founder, I DO Organization, India:

- An organization aimed at synergizing the efforts of different social sector organizations.
- Led the planning and execution of "Champions of Change program" – in which young people were placed with different social sector organizations; later collated their experiences and learnings.

Jul, 2013: Agumbe Rainforest Research Station, Karnataka, India:

- Studied road kills of amphibians and reptiles near a reserve forest and agricultural fields.
- Studied bush frog populations in disturbed and undisturbed areas.

2010-2014: Nirmaan Organization, Pilani Chapter, India:

- 2011-2012: Project Leader, School Adoption Program: Helped introduce new teaching pedagogies in a govt school; Facilitated construction of shelter in the school
- 2012-2013: Education Operations Head: Initiated Utkarsh Scholarship & Mentorship Program; Coordination and information exchange between 5 education projects

SKILLS

- *Statistics* | Regression analysis, Mixed models, Time-series analysis, Multivariate analysis, Spatial and Temporal Autocorrelation structures
- *Programming and Software* | R Programming Language, QGIS, Google Earth Engine, GitHub
- *Research Methodologies* | Experimental designs, Vegetation monitoring, Social science surveys (qualitative and quantitative)
- *Laboratory analysis* | Biological and chemical laboratory analysis of soil, water, and plant samples
- *Leadership and Mentoring* | Leading teams in both scientific and social contexts, Guiding MSc Students
- *Photography*